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Machine Learning (911.236), **Statistical Learning Theory** (536.505), **Advanced Machine Learning** (911.936)

Exercise sheet **E**

Exercise 1. 6 P.

Let $u \in \mathbb{R}^d \setminus \{0\}$ be fixed. Consider the hypothesis class

$$\mathcal{H} = \{x \mapsto \mathbf{1}[\langle u, x\rangle + b \geq 0] : b \in \mathbb{R}\} \quad ,$$

i.e., any label is in $\{0, 1\}$ and the decision boundary is a hyperplane with normal vector u. For a finite set $S = \{x_1, \ldots, x_n\} \subset \mathbb{R}^d$, define

$$t_i = \langle u, x_i\rangle, \qquad i = 1, \ldots, n \quad .$$

A labeling $y_1, \ldots, y_n \in \{0, 1\}$ is called *projection-monotone* if for all $i, j \in \{1, \ldots, n\}$ we have

$$t_i \leq t_j \quad \Longrightarrow \quad y_i \leq y_j \quad .$$

Tasks:

(a) Show that every labeling realizable by $\mathcal{H}$ on S is projection-monotone.

(b) Assume that the values $t_1, \ldots, t_n$ are pairwise distinct. Show that every projection-monotone labeling of S is realizable by some hypothesis in $\mathcal{H}$.

(c) Deduce that the growth function of $\mathcal{H}$ satisfies

$$\tau_{\mathcal{H}}(n) \leq n + 1 \quad .$$

(d) Determine the VC dimension of $\mathcal{H}$.

(e) Lets say we would claim

> "Since the input space is $\mathbb{R}^d$ and the decision boundary is a hyperplane (with bias), the VC dimension of $\mathcal{H}$ should be $d + 1$."

Explain precisely why this argument is incorrect.

Total #points: 6 P.